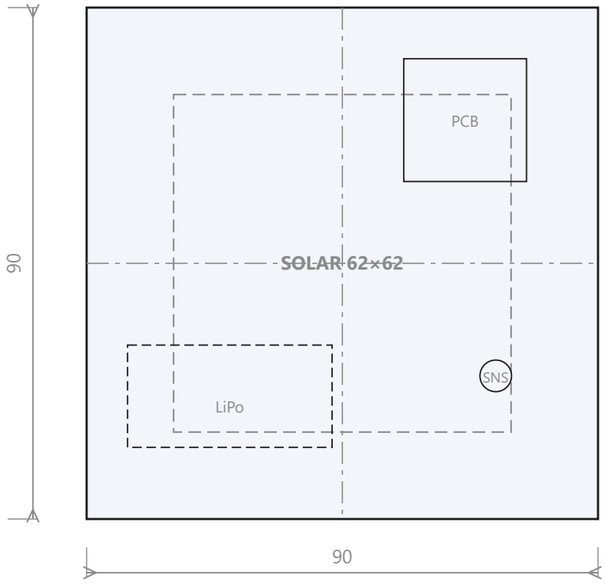
S · MITE
GENERAL ARRANGEMENT — SWARM UNIT

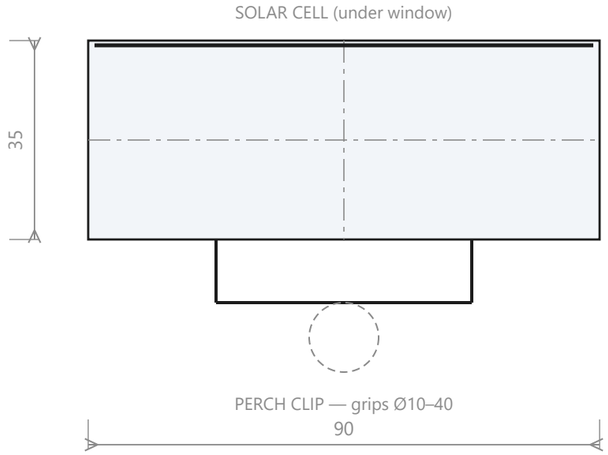
AF-S-101
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FOR REVIEW

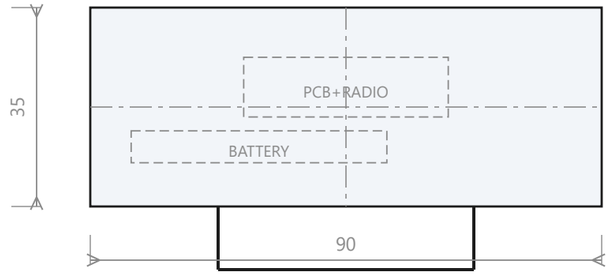
TOP VIEW



FRONT VIEW



SIDE VIEW



WHAT THIS NODE IS

— One member of the fire-watch swarm — the swarm IS the sensor, not the individual bee.
— Minimal on purpose: solar top, LoRa radio, temp/RH/VOC sensing, perch clip.
— Default state = perched and reporting. Flight is a later role, not the power budget.
— Same signed mesh identity as every ArcticChain node.

SWARM ROLES (ASSIGNED BY MESH AI)

ROLE	PAYLOAD	JOB
Sensor-bee	temp/RH + VOC	sniff anomalies — primary detector
Eye-bee	thermal/visual (rev-C)	converge on fixes to confirm
Relay-bee	LoRa only, high perch	extend range over ridges
Charger-bee	dock contacts	perch at Anchor, top off, rebroadcast

GEN-1 VS ROADMAP

STAGE	WHAT	GATE
Gen-1 (now)	~90 mm puck, solar+LoRa+gas, perch clip	orderable parts only
Gen-2	smaller, printed antenna, lower cost	volume + field data
Gen-3 (R&D)	sub-bee perched motes, degradable	transient-electronics — see D SEED

ARCTICCHAIN — RURAL NODE FLEET
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DRAWN **Surafel Negussie**
DATE **2026-11-12**
SCALE **NTS**
SHEET **1/3**
DWG NO **AF-S-101**
UNITS **mm**

PARTS LIST — GEN-1 (ALL ORDERABLE IN CANADA)

#	PART	SUPPLIER	APPROX CAD	QTY
1	ESP32-S3-DevKitC-1 / S3-WROOM carrier	PiShop / DigiKey	$8–15	1
2	EBYTE E22-900M30S SX1262 LoRa 915 MHz	DFRobot / AliExpress	$15–25	1
3	Sensirion ENS160 VOC/AIQ (I2C)	DigiKey / Mouser	$6–10	1
4	Sensirion SHT41 temp/RH (I2C)	DigiKey / Mouser	$4–7	1
5	60×60 mm ~1 W 5 V poly solar cell	Amazon.ca / supplier	$5–10	1
6	503450 LiPo 2000 mAh + JST-PH	Amazon.ca / RC supplier	$12–18	1
7	CN3791 MPPT solar charge module	AliExpress / Amazon.ca	$3–6	1
8	Custom carrier PCB ~38×38 mm (JLCPCB)	JLCPCB / PCBWay	$15–30	1
9	Printed enclosure set + spring strip	Print farm	$5–10	1

POWER BUDGET (ESTIMATES — VERIFY ON BENCH)

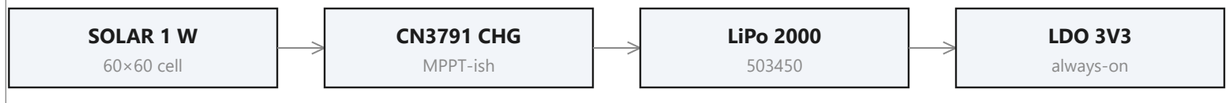
STATE	CURRENT	DUTY	ENERGY/DAY
Deep sleep (ESP32)	~10 µA	~95%	~0.003 Wh
Sense + process	~80 mA @3.3 V	~4%	~0.013 Wh
LoRa TX burst	~120 mA @3.3 V	~1%	~0.010 Wh
RX listen windows	~15 mA	~5%	~0.006 Wh
TOTAL			≈0.03 Wh/day

2000 mAh LiPo ≈ 7.4 Wh → ~200 days theoretical; winter + self-discharge → plan 60–90 day autonomy.

CARRIER PCB PIN MAP

NET	ESP32-S3 GPIO	FUNCTION
LORA_SCK/MISO/MOSI	12 / 13 / 11	SX1262 SPI
LORA_NSS/RST/BUSY	10 / 9 / 8	chip select · reset · DIO1 IRQ
I2C_SDA/SCL	1 / 2	ENS160 (0x53) + SHT41 (0x44)
BATT_ADC	4	battery voltage sense
POGO_CHG	3	dock charge detect
SOLAR_OK	5	charge status

SYSTEM BLOCK



ESP32-S3 + SX1262 + ENS160 on 38×38 carrier; sensors on I2C; radio on private SPI.

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SHEET	**2/3**
DATE	**2026-11-12**
DWG NO	**AF-S-102**
SCALE	**NTS**
UNITS	**mm**

ASSEMBLY SEQUENCE

1. Print shell, lid, battery bay, clip — ASA/PC-ABS shell, PETG clip.
2. Populate carrier PCB: ESP32-S3, SX1262, ENS160, SHT41, charge circuit.
3. Seat battery, connect JST; verify charge path (bench supply or solar sim).
4. Close lid — gasket bead, solar under window film.
5. Attach perch clip; bench-test join to test mesh.
6. Field trial: perch ×3 units on fence line, log 7 days.

FIRMWARE

ITEM	DETAIL
Runtime	ESP-IDF (C) gen-1; Rust port when mesh-net envelope lands
Envelope	same signed format as fleet — MUST match mesh-net
Duty cycle	deep-sleep 95% · sample 5 min · TX on anomaly/schedule
Data	temp/RH/VOC + battery + RSSI + role → <200 B frame
OTA	signed updates over mesh (rev-C); USB/SD bench flash

ROLE ASSIGNMENT (SKETCH)

— All Miters broadcast hello + battery + capability every N hours.
— Anchor/Brick AI computes coverage gaps → assigns roles via signed cmd.
— Sensor-bees concentrate near fuel breaks/ridgelines in fire season.
— Charger-bees migrate to Anchor docks when battery <20%.
— Roles are soft state — units self-organize if coordinator is gone.

ACCEPTANCE / TEST CRITERIA

TEST	PASS CONDITION
Mesh range	signed observations A→B via Brick/Anchor ≥3 km LOS
Autonomy	≥60 days at 1 sample/10 min + hourly beacon
Cold	operates -30 °C; solar resumes charge at -10 °C
Perch	clip holds Ø25 mm branch through wind-shake test
False positive	<1 smoke alarm/week/unit in dust/inversion

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SHEET **3/3**
DATE **2026-11-12**
DWG NO **AF-S-103**
SCALE **NTS**
UNITS **mm**